

ADVANCED NETWORK SECURITY AND ARCHITECTURES

METI

Report Module 2 [EN]

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1 Introduction

2 IKEv1 with Pre-Shared Keys

3 IKEv1 with Digital Signatures

3.1 Introduction

This section describes the configuration and analysis of IPSec VPN using IKEv1 with digital certificates (PKI-based authentication using SCEP). The objective is to replace pre-shared keys with RSA signatures issued by a Certification Authority (CA).

3.2 PKI Installation Verification

The Certification Authority (CA) and router certificates were verified using IOS show commands.

3.2.1 CA Configuration and Certificate (RA)

The CA is configured on router RA and operates in automatic enrollment mode.

```
RA#show crypto pki server
Certificate Server myCA:
  Status: enabled
  State: enabled
  Server's configuration is locked (enter "shut" to unlock it)
  Issuer name: cn=saarCA
  CA cert fingerprint: 46CB9C10 CC370311 8CFCB545 E4DD2F58
  Granting mode is: auto
  Last certificate issued serial number (hex): 1
  CA certificate expiration timer: 16:17:23 UTC May 14 2029
  CRL NextUpdate timer: 22:17:23 UTC May 15 2026
  Current primary storage dir: nvram:
  Database Level: Minimum - no cert data written to storage
```

The CA certificate installed on RA confirms that RA is acting as a trusted root authority:

```
RA#show crypto ca certificate
CA Certificate
  Status: Available
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end date: 16:17:23 UTC May 14 2029
  Associated Trustpoints: myCA
  Storage: nvram:saarCA#1CA.cer
```

The detailed certificate view confirms RSA key usage and X.509 extensions:

```
RA#show crypto ca certificate verbose
CA Certificate
  Status: Available
  Version: 3
  Certificate Serial Number (hex): 01
  Certificate Usage: Signature
  Issuer:
    cn=saarCA
  Subject:
    cn=saarCA
  Validity Date:
    start date: 16:17:23 UTC May 15 2026
    end   date: 16:17:23 UTC May 14 2029
  Subject Key Info:
    Public Key Algorithm: rsaEncryption
    RSA Public Key: (1024 bit)
  Signature Algorithm: MD5 with RSA Encryption
  Fingerprint MD5: 46CB9C10 CC370311 8CF5B545 E4DD2F58
  Fingerprint SHA1: 0B6BF47E F0932FC1 54A33FA8 4C1F48D1 80083958
  X509v3 extensions:
    X509v3 Key Usage: 86000000
      Digital Signature
      Key Cert Sign
      CRL Signature
    X509v3 Subject Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
    X509v3 Basic Constraints:
      CA: TRUE
    X509v3 Authority Key ID: 42BD5391 0D07D957 6CC26630 AABA09E4 A93162E2
    Authority Info Access:
  Associated Trustpoints: myCA
  Storage: nvram:saarCA#1CA.cer
```

3.2.2 Router Certificates (R1)

R1 successfully enrolled with the CA and received both the CA certificate and its own identity certificate:

```
R1#show crypto pki certificates
Certificate
  Status: Available
  Certificate Serial Number (hex): 03
  Certificate Usage: General Purpose
  Issuer:
    cn=saarCA
```

Subject:
Name: R1
Serial Number: 9S08DYF8WAKSOLHEG3ZVL
hostname=R1+serialNumber=9S08DYF8WAKSOLHEG3ZVL
Validity Date:
start date: 16:33:16 UTC May 15 2026
end date: 16:33:16 UTC May 15 2027
Associated Trustpoints: myTrustpoint

The CA certificate is also stored locally to validate the trust chain:

CA Certificate
Status: Available
Certificate Serial Number (hex): 01
Certificate Usage: Signature
Issuer:
cn=saarCA
Subject:
cn=saarCA
Validity Date:
start date: 16:17:23 UTC May 15 2026
end date: 16:17:23 UTC May 14 2029
Associated Trustpoints: myTrustpoint

3.2.3 Summary

The outputs confirm that:

- RA is correctly configured as a trusted CA (myCA).
- R1 has successfully obtained a signed X.509 certificate from the CA.
- The certificate chain is self-signed at the CA level.
- RSA key pairs are associated with each identity for IKEv1 authentication.

3.3 SCEP Enrollment Analysis

SCEP (Simple Certificate Enrollment Protocol) is used to automate certificate distribution between routers and the CA using HTTP.

The enrollment process consists of:

- CA certificate request
- CA certificate response
- Router certificate request
- Router certificate issuance

The captured HTTP messages show certificate exchange between R1 and the CA. The content-type field indicates certificate payload transfer (.crt).